

Aastha Bajpai

Uttar Pradesh, India

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| [LinkedIn](#)

| [GitHub](#)

| [Portfolio](#)

EDUCATION

National Institute of Technology (NIT), Jamshedpur

Master of Computer Applications (MCA), CGPA: 7.75

Aug 2024 – May 2027

Jamshedpur, Jharkhand

Chhatrapati Shahu Ji Maharaj University (CSJMU)

Bachelor of Computer Applications (BCA), CGPA: 7.56

Aug 2019 – May 2022

Kanpur, Uttar Pradesh

TECHNICAL SKILLS

Languages: C++, JavaScript (ES6+), SQL

Frontend: React.js, HTML5, CSS3

Backend: Node.js, Express.js, REST APIs, JWT Authentication, Socket.IO

Databases: MongoDB, MySQL

CS Fundamentals: Data Structures & Algorithms, OOPS, DBMS, Operating Systems, Computer Networks

Tools & Platforms: Git, GitHub, Postman, VS Code, Vercel, Render

PROJECTS

AI-Integrated College ERP System

[Live](#) | [GitHub](#)

React.js, Node.js, Express.js, MongoDB, JWT, Socket.IO, Gemini AI: A role-based ERP platform unifying admissions, fee collection, hostel allocation, and exam records into one system.

- Engineered **15+ RESTful APIs** with JWT-based Role-Based Access Control, powering dedicated student, staff, and admin portals and removing redundant data entry across **5+ workflows**.
- Built a real-time hostel occupancy dashboard with **Socket.IO**, cutting manual allocation queries by **~60%**.
- Integrated an AI-powered **Student Performance Predictor** using the Gemini AI API to flag at-risk students early.
- Automated digital fee receipts, reducing processing time by **~40%** over manual ledger workflows.

BhashaShikho

[Live](#) | [GitHub](#)

React.js, Node.js, Express.js, MongoDB, JWT, Zustand, Stream: A real-time language exchange platform enabling chat and video-based peer learning.

- Built **1-on-1 and group video calls** with screen sharing and session recording across **32 UI themes**.
- Secured user sessions with **JWT authentication**; managed state via **Zustand** and server sync via **TanStack Query**.
- Implemented real-time chat with typing indicators using the **Stream SDK**, achieving sub-100ms message delivery.

House Price Prediction System

[Live](#) | [GitHub](#)

Python, Scikit-learn, Pandas, NumPy, Streamlit: An ML web app predicting house prices from key property features.

- Built an end-to-end pipeline with feature encoding, **StandardScaler** normalization, and **Linear Regression** on **500+ records**, achieving a strong R^2 score.
- Deployed an interactive **Streamlit** app returning instant predictions with MAE, MSE, RMSE, and R^2 metrics.

ACHIEVEMENTS & CODING PLATFORMS

- NIMCET 2024** — **All India Rank 519** among ~25,000 appearing candidates nationwide, securing admission to NIT Jamshedpur's MCA programme.
- Postman API Fundamentals Student Expert** — Certified by Postman (Jul 2025) for proficiency in building, testing, and documenting REST APIs.
- HackerRank** — Earned **5-Star ratings** in C++ and SQL, reflecting strong command over algorithms and database querying.
- LeetCode** — Solved **200+ problems** spanning arrays, trees, graphs, and dynamic programming.
- GeeksforGeeks** — Solved **60+ problems**, strengthening core data structures and algorithmic fundamentals.

EXTRACURRICULAR

Technical Coordinator — Society of Computer Science & Engineering (SCSE)

Feb 2025 – Feb 2026

NIT Jamshedpur

- Organized **3+ coding competitions** and hackathons as Technical Coordinator, driving participation from **80+ students**.

Creative & Content Coordinator — Entrepreneurship Cell (E-Cell)

Sep 2024 – Dec 2025

NIT Jamshedpur

- Led digital content and visual branding for **5+ E-Cell events**, including summits attended by **150+ students**.